AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2019

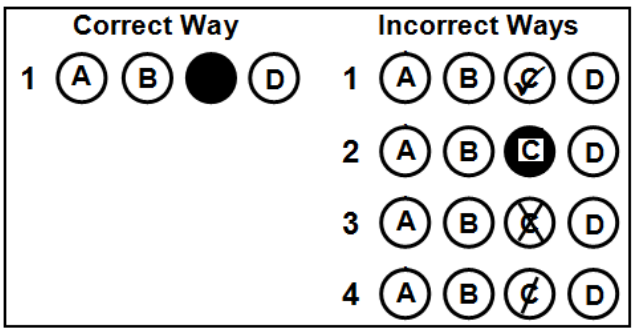
Biology Paper I

Time: 45 minutes Marks: 30

April/ May 2019 for Teaching & Learning only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. As compared to the nephrons of fresh water vertebrates, the nephrons of desert rodents have

   A. longer loop of Henle.
   B. shorter collecting duct.
   C. less convoluted distal tubule.
   D. more convoluted proximal tubule.

2. Vasa recta is the cluster of blood capillaries present in the nephron of human beings. It is derived from **X** and surrounds **Y**, where **X** are arterioles and **Y** are tube-like structures.

   Which of the following CORRECTLY identifies **X** and **Y**?

	X	**Y**
A	Efferent arterioles	Proximal convoluted tubules
B	Afferent arterioles	Proximal convoluted tubules
C	Efferent arterioles	Loop of Henle
D	Afferent arterioles	Loop of Henle

3. A person is suffering from hyper-secretion of aldosterone.

   The laboratory report of his urine composition will show the absence or very low level of

   A. glucose.
   B. creatinine.
   C. sodium ions.
   D. phosphate ions.

4.
   I. Globular proteins
   II. Thin thread-like structure
   III. Wound around the actin filament

   The feature(s) which describe(s) tropomyosin molecules is/ are

   A. I only.
   B. III only.
   C. I and II.
   D. II and III.

5. In the muscle fibre, transverse tubules (T-tubules) are formed by the invagination of sarcolemma. The function of these T-tubules is to

   A. unblock the binding sites of actin.
   B. conduct nerve impulse into the cell.
   C. synthesise glycogen in the sarcoplasm.
   D. store calcium ions in the sarcoplasmic reticulum.

April/ May 2019
for
Teaching & Learning only

6. The animal that uses jet propulsion to locomote is

   A. snail.
   B. star fish.
   C. jelly fish.
   D. earthworm.

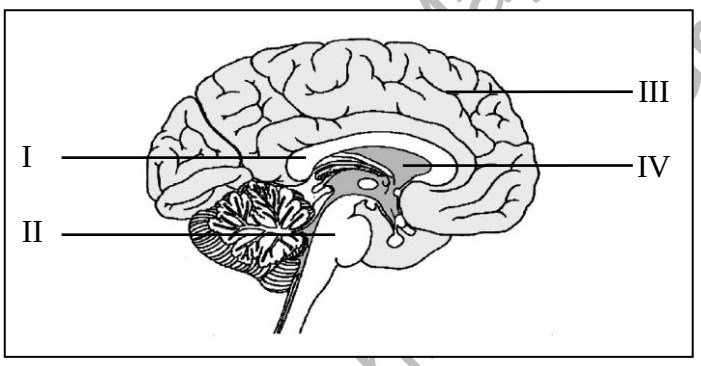
7. Auxins and gibberellins are plant hormones. The common role of these hormones is to

   A. delay both bud initiation and leaf senescence.
   B. promote both bud initiation and leaf senescence.
   C. promote bud initiation and delay leaf senescence.
   D. promote leaf senescence and delay bud initiation.

8. In contrast to motor and relay neurons, the sensory neurons contain

   A. single axon.
   B. single dendron.
   C. multiple dendrites.
   D. multiple nodes of Ranvier.

9. The given diagram shows a section through the midline of the human brain.



   The labelled structure that connects right and left cerebral hemisphere is

   A. I.
   B. II.
   C. III.
   D. IV.

10. In human beings, what is the ratio of the number of gametes produced from one male primary sex cell to the number of gametes produced from one female primary sex cell?

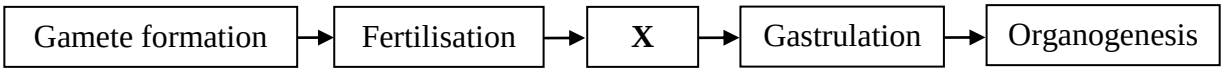
    A. 1 : 3
    B. 1 : 4
    C. 3 : 1
    D. 4 : 1

Teaching and learning only May 2019

11. The phase of uterine cycle in which degeneration of endometrium occurs is

A. luteal.
B. follicular.
C. ovulation.
D. menstruation.

12. Following are some of the events that occur during the development of animals.



The structures formed during the event **X** are

A. somites.
B. blastomeres.
C. epiblast and hypoblast.
D. neural groove and neurocoel.

13. The genetic code is universal across organisms.

This feature is advantageous in the process of

A. karyotyping.
B. amplification of a gene.
C. isolation of genes from organism.
D. synthesis of insulin from bacteria.

14. A section of deoxyribonucleic acid (DNA) contains the following sequence of bases.

G A T C A G C C A T A C

The number of amino acids that the given section of DNA can code is

A. 3.
B. 4.
C. 6.
D 12.

15. Which of the following CORRECTLY represents the separation of sister chromatids and homologous chromosomes during mitosis and meiosis?

	Mitosis	**Meiosis I**	**Meiosis II**
A	Sister chromatids	Sister chromatids	Homologous chromosomes
B	Homologous chromosomes	Homologous chromosomes	Sister chromatids
C	Sister chromatids	Homologous chromosomes	Sister chromatids
D	Homologous chromosomes	Sister chromatids	Homologous chromosomes

April/May 2019 for Teaching & Learning only

16. A cell contains 8 chromosomes in its diploid state.

When this cell undergoes meiosis, its daughter cells would contain

A. two genetically identical chromosomes each.
B. two genetically different chromosomes each.
C. four genetically identical chromosomes each.
D. four genetically different chromosomes each.

17. When two heterozygotes (AaBb) of completely dominant traits are crossed, then the probability of appearance of AABB genotype would be

A. 1/16
B. 3/16
C. 6/16
D. 9/16

18. In a particular type of pigeons, brown (B) feather and red (R) eye colour is completely dominant over white (b) feather and black (r) eye colour.

The following result is obtained from a cross between two such pigeons.

1	:	1	:	1	:	1
BbRr		Bbrr		bbRr		bbrr

The genotypes of the parent pigeons would be

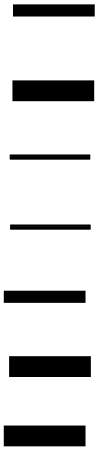
A. BBrr and Bbrr.
B. Bbrr and bbRr.
C. BBRr and Bbrr.
D. BbRr and bbRR.

19. The genotypes of parents of a baby girl who is born with haemophilia would be

	Genotype of Mother	**Genotype of Father**
A	carrier	normal
B	haemophiliac	normal
C	carrier	haemophiliac
D	normal	haemophiliac

April/ May 2019 for Teaching & Learning only

20. The given table shows a forensic investigation result comprising of DNA fingerprints.

DNA Fingerprints of Blood Sample from				
Crime Scene	**Suspect 1**	**Suspect 2**	**Suspect 3**	**Suspect 4**
				

Based on the given result, the culprit would be

A. suspect 1.
B. suspect 2.
C. suspect 3.
D. suspect 4.

21. A 16 weeks pregnant woman, who already has a son with haemophilia, visits a genetic counsellor to know about the genetic health of her second baby.

The diagnostic test that the genetic counsellor refers to the woman would be

A. ultrasound.
B. gene therapy.
C. tissue culture.
D. amniocenteses.

22. One of the hypotheses about origin of life is that life began deep in the oceans, in underwater hot springs. The occurrence of which of the following organisms supports the given hypothesis?

A. Sycon
B. Green algae
C. Zooflagellates
D. Archaebacteria

April/ May 2
for
Teaching & Learning

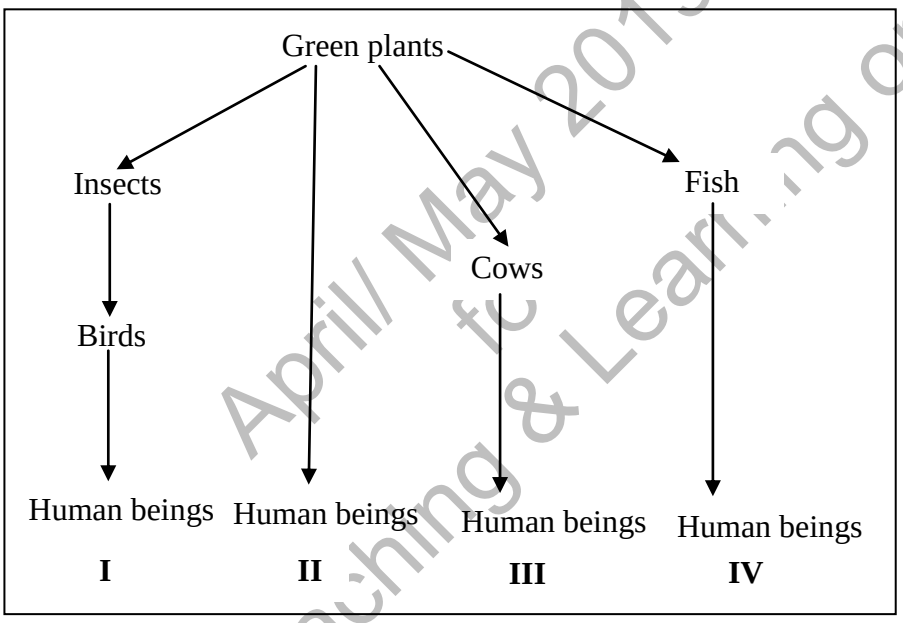
23. According to Hardy-Weinberg law, the frequencies of alleles and genotypes remain constant in a population across generations when

A. mutation takes place.
B. migration takes place.
C. random mating takes place.
D. natural selection takes place.

24. Genetic drift MOST likely occurs when the population's

A. size is very small.
B. gene pool is diverse.
C. rate of reproduction is high.
D. species are protected and conserved.

25. The given diagram shows four food chains.



The food chain that transfers MINIMUM energy to the human beings is

A. I.
B. II.
C. III.
D. IV.

26. In a pyramid of energy, if the energy at a specific trophic level is 1,000 J, the amount of energy available for the next higher trophic level would be

A. 10 J.
B. 100 J.
C. 1,000 J.
D. 10,000 J.

April/ May 2019 for Teaching & Learning only

27. The terrestrial biome in which the average temperature remains low throughout the year is

A. tundra.
B. grassland.
C. tropical rain forest.
D. temperate deciduous forests.

28. The reason that coniferous forests mainly comprise of evergreen coniferous trees is

A. low altitude.
B. heavy rainfall.
C. highly fertile soil.
D. short summer season.

29. Soil moisture is limited in the grassland ecosystem because of

	Precipitation	**Evaporation**
A	low	high
B	low	low
C	high	low
D	high	high

30. The ideal place for the conservation and protection of snow leopard is

A. zoo garden.
B. national park.
C. botanical garden.
D. wildlife corridors.

April/ May 2019 for Teaching & Learning only

END OF PAPER

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only

Please use this page for rough work

April/ May 2019
for
Teaching & Learning only